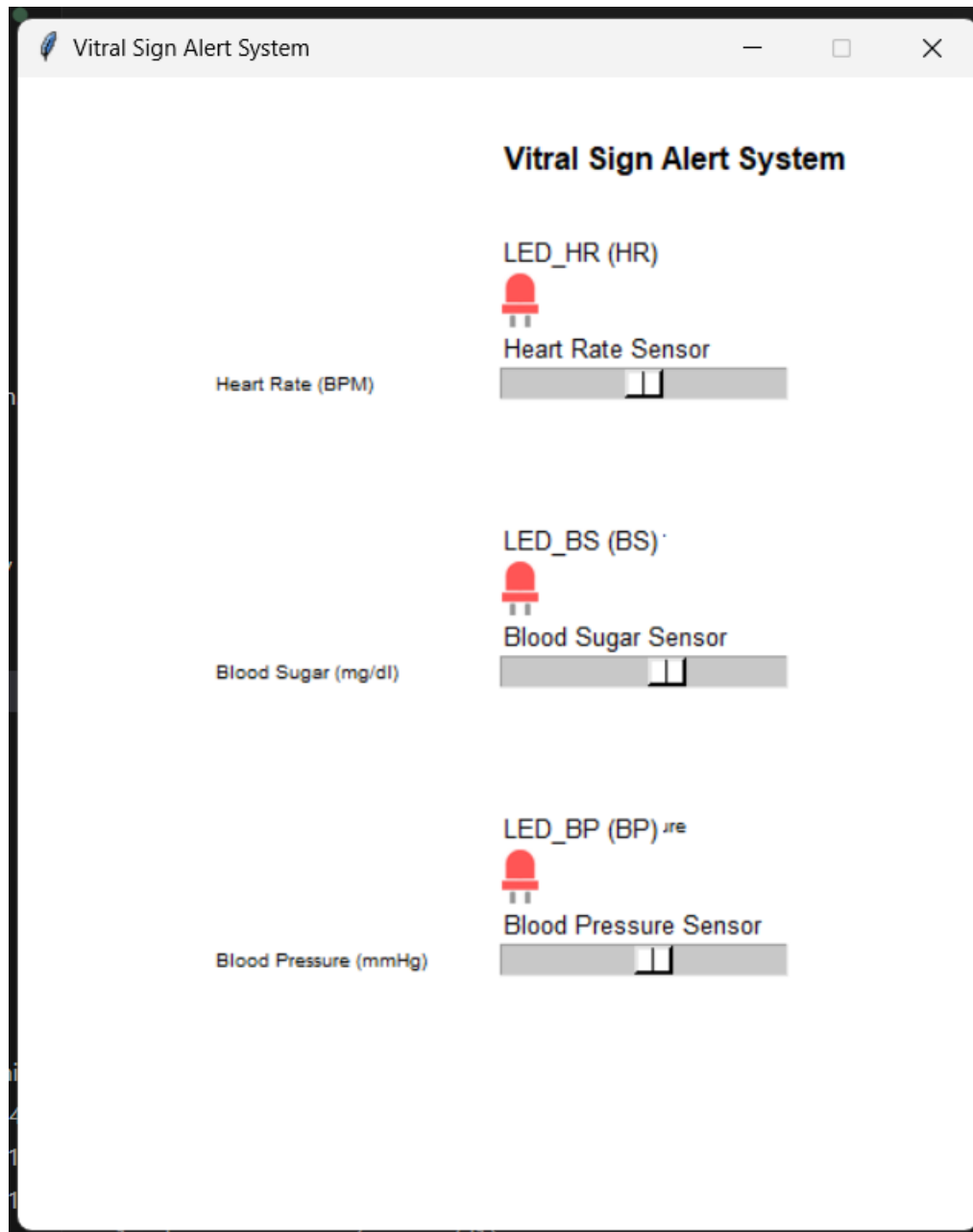




email_sending.py

```
import smtplib
import email.mime.multipart
from email.mime.text import MIMEText

user_name = "Thanawat Lhuangwirot"
user_email = 'thanawatlhuangwirot@gmail.com'
gmail_password = 'pjaw cufa bgax xfqj'
target_name = "Thanawat Lhuangwirot"
target_email = "thanawat.lue@gpo.or.th"
message = "This is the body of the Email Test message."
```

```

def send_email(user_name, user_email, gmail_password, target_name, target_email, message):

    try:
        msg = email.mime.multipart.MIMEMultipart()
        msg['to'] = f'{target_name}<{target_email}>'
        msg['from'] = f'{user_name}<{user_email}>'
        msg['subject'] = 'Email Test'
        msg.add_header('reply-to', user_email)
        msg.attach(MIMEText(message, 'plain'))
        session = smtplib.SMTP('smtp.gmail.com', 587)
        session.starttls()
        session.login(user_email, gmail_password)
        message = msg.as_string()
        session.sendmail(user_email, target_email, message)
        session.quit()
        print('Email sent.')
    except Exception as e:
        print('Email failed to send.')
        print(e)

# send_email(user_name, user_email, gmail_password, target_name, target_email, message)

```

vitral_sign.py

```

from gpiozero import Servo, MCP3008, Button, LED, Buzzer
from time import sleep
import signal
from email_sending import send_email

user_name = "Thanawat Lhuangwirot"
user_email = 'thanawatlhuangwirot@gmail.com'
gmail_password = 'pjaw cufa bgax xfqj'
target_name = "Thanawat Lhuangwirot"
target_email = "thanawat.lue@gpo.or.th"

print("Vital Sign Alert System is running...")

led_HeartRate = LED(17)
led_BloodSugar = LED(27)
led_BloodPressure = LED(22)
HeartRate_sensor = MCP3008(channel = 0)
BloodSugar_sensor = MCP3008(channel = 1)
BloodPressure_sensor = MCP3008(channel = 2)

hr_alert_sent = False
bs_alert_sent = False
bp_alert_sent = False

```

```

email_cooldown = 120 # ส่งทุกๆ 1 นาที 120 * 0.5

hr_cooldown = 0
bs_cooldown = 0
bp_cooldown = 0

def main():
    global hr_alert_sent, bs_alert_sent, bp_alert_sent
    global hr_cooldown, bs_cooldown, bp_cooldown

    print("System active. Press Ctrl+C to exit.")

    while True:
        # อัปเดตตัวนับ cooldown
        if hr_cooldown > 0:
            hr_cooldown -= 1
        if bs_cooldown > 0:
            bs_cooldown -= 1
        if bp_cooldown > 0:
            bp_cooldown -= 1

        # HeartRate_sensor (if more than 120 bpm, send email notify user, max 300 bpm)
        HR = HeartRate_sensor.value
        print(f"Heart Rate: {HR:.2f} ({int(HR * 300)} BPM)")

        if HR >= 0.4:
            led_HeartRate.on()
            if not hr_alert_sent and hr_cooldown == 0:
                HR_message = f"ALERT: High Heart Rate detected! Current value: {int(HR * 300)} BPM ({HR:.2f})"
                send_email(user_name, user_email, gmail_password, target_name, target_email,
                    HR_message)
                hr_alert_sent = True
                hr_cooldown = email_cooldown
                print("Heart Rate alert ")
            else:
                led_HeartRate.off()
                if hr_alert_sent and hr_cooldown == 0:
                    # HR_normal_message = f"NORMAL: Heart Rate returned to normal: {int(HR * 300)} BPM ({HR:.2f})"
                    # send_email(user_name, user_email, gmail_password, target_name, target_email,
                    HR_normal_message)
                    hr_alert_sent = False
                    hr_cooldown = email_cooldown
                    # print("Heart Rate normal email sent!")

        # BloodSugar_sensor (if more than 200 mg/dl, send email notify user, max 400 mg/dl)
        BS = BloodSugar_sensor.value
        print(f"Blood Sugar: {BS:.2f} ({int(BS * 400)} mg/dl)")

```

```

if BS >= 0.5:
    led_BloodSugar.on()
    if not bs_alert_sent and bs_cooldown == 0:
        BS_message = f"ALERT: High Blood Sugar detected! Current value: {int(BS * 400)} mg/dl
({BS:.2f})"
        send_email(user_name, user_email, gmail_password, target_name, target_email,
BS_message)
        bs_alert_sent = True
        bs_cooldown = email_cooldown
        print("Blood Sugar alert email sent!")
    else:
        led_BloodSugar.off()
        if bs_alert_sent and bs_cooldown == 0:
            # BS_normal_message = f"NORMAL: Blood Sugar returned to normal: {int(BS * 400)}
mg/dl ({BS:.2f})"
            # send_email(user_name, user_email, gmail_password, target_name, target_email,
BS_normal_message)
            bs_alert_sent = False
            bs_cooldown = email_cooldown
            # print("Blood Sugar normal email sent!")

# BloodPressure_sensor (if more than 140 mmHg, send email notify user, for max 300 mmHg)
BP = BloodPressure_sensor.value
print(f"Blood Pressure: {BP:.2f} ({int(BP * 300)} mmHg)")

if BP >= 0.47:
    led_BloodPressure.on()
    if not bp_alert_sent and bp_cooldown == 0:
        BP_message = f"ALERT: High Blood Pressure detected! Current value: {int(BP * 300)}
mmHg ({BP:.2f})"
        send_email(user_name, user_email, gmail_password, target_name, target_email,
BP_message)
        bp_alert_sent = True
        bp_cooldown = email_cooldown
        print("Blood Pressure alert email sent!")
    else:
        led_BloodPressure.off()
        if bp_alert_sent and bp_cooldown == 0:
            # BP_normal_message = f"NORMAL: Blood Pressure returned to normal: {int(BP * 300)}
mmHg ({BP:.2f})"
            # send_email(user_name, user_email, gmail_password, target_name, target_email,
BP_normal_message)
            bp_alert_sent = False
            bp_cooldown = email_cooldown
            # print("Blood Pressure normal email sent!")

sleep(0.5)

if __name__ == "__main__":
    main()

```

vitral_sign_simulation.py

```
from tkgpio import TkCircuit

configuration = {
    "name": "Vitral Sign Alert System",
    "width": 500,
    "height": 600,

    "leds": [
        {"x": 250, "y": 100, "name": "LED_HR (HR)", "pin": 17},
        {"x": 250, "y": 250, "name": "LED_BS (BS)", "pin": 27},
        {"x": 250, "y": 400, "name": "LED_BP (BP)", "pin": 22}
    ],

    "adc": {
        "mcp_chip": 3008,
        "potentiometers": [
            {"x": 250, "y": 150, "name": "Heart Rate Sensor", "channel": 0},
            {"x": 250, "y": 300, "name": "Blood Sugar Sensor", "channel": 1},
            {"x": 250, "y": 450, "name": "Blood Pressure Sensor", "channel": 2}
        ]
    },

    "labels": [
        {"x": 250, "y": 30, "text": "Vitral Sign Alert System", "font": ("Arial", 16, "bold")},
        {"x": 250, "y": 80, "text": "High Heart Rate", "font": ("Arial", 10)},
        {"x": 250, "y": 230, "text": "High Blood Sugar", "font": ("Arial", 10)},
        {"x": 250, "y": 380, "text": "High Blood Pressure", "font": ("Arial", 10)},
        {"x": 100, "y": 150, "text": "Heart Rate (BPM)", "font": ("Arial", 10)},
        {"x": 100, "y": 300, "text": "Blood Sugar (mg/dl)", "font": ("Arial", 10)},
        {"x": 100, "y": 450, "text": "Blood Pressure (mmHg)", "font": ("Arial", 10)}
    ]
}

circuit = TkCircuit(configuration)

@circuit.run
def main():
    import vitral_sign

    vitral_sign.main()
```

E-mail inbox:

Heart rate

The screenshot shows an email client interface. On the left is a sidebar with a 'Favourites' section containing 'ธนวัฒน์ เหลืองวิโรจน์' and an 'Inbox' with 6 items. Below this are 'Drafts', 'Sent Items', 'Deleted Items', 'Archive', 'Chem_Lab' (1 item), and 'Conversation History'. The main area is divided into two panes. The left pane shows the 'Inbox' list with three emails from 'Thanawat Lhuangwirot' at 11:28, 11:27, and 11:27. The first email is selected. The right pane shows the selected email, 'Email Test', from 'Thanawat Lhuangwirot <thanawatlhuangwirot@gmail.com>' at 11:27. The email body contains the text: 'ALERT: High Heart Rate detected! Current value: 149 BPM (0.50)'.

Blood Sugar

This screenshot is similar to the previous one but shows a different email. The sidebar is identical. The selected email in the 'Inbox' list is the one from 11:27 with the subject 'Email Test'. The email body in the right pane reads: 'ALERT: High Blood Sugar detected! Current value: 235 mg/dl (0.59)'.

Blood Pressure

This screenshot shows the same email client interface with a different email selected. The sidebar remains the same. The selected email in the 'Inbox' list is the one from 11:28 with the subject 'Email Test'. The email body in the right pane reads: 'ALERT: High Blood Pressure detected! Current value: 161 mmHg (0.54)'.